

FIN ST912–ST1001

What Prevents Nadsoliton Annihilation?

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No nonlinear/topological/quantum persistence mechanism is silently assumed to be strict-derived.

Abstract

ST912–ST1001 investigate what, if anything, prevents the FIN nadsoliton from annihilating. Four inequivalent meanings are separated: extinction of the entire state, delocalization of its pattern, loss of coherence, and loss of topological/attractor identity.

For the frozen strict generator, closed unitary dynamics preserves Hilbert norm and makes the zero state unreachable from nonzero initial data. Heat dynamics preserves total mass but exponentially erases every mean-zero pattern at the spectral-gap rate $0.754121\dots$, converging to the uniform state. A second-order wave model preserves phase-space energy. Dephasing destroys coherence while preserving spectral populations. These theorems protect conserved totals, not the identity of a localized nonlinear object.

Conditional stronger mechanisms are classified. A $U(1)$ Noether charge prevents global zero when positive. A discrete winding is homotopy-protected only while all amplitudes remain nonzero and phase increments avoid their branch cut; linear unitary dynamics does not guarantee that gap. DNLS conserves norm and energy but a FIN-specific localized orbitally stable breather is unproved. Pump/loss can create a nonzero attracting norm, but persistence consumes a nonequilibrium resource and disappears when the pump is removed. Superselection charge, dark states, decoherence-free subspaces and quantum error correction can protect quantum information, but require additional charge/noise/code/recovery structures and often an entropy sink.

The current FIN model contains no multisoliton collision law, so annihilation in a collision is not yet a well-posed internal question. The final result is a no-go for strict nadsoliton identity protection: current strict mathematics protects unitary norm, heat mass and conditional wave energy, but does not prove localization, topological charge, pumping, quantum code, orbital stability or collision balance. A conditional persistence theory is supplied, together with pump-off, winding, perturbation, noise and information-recovery tests; no laboratory apparatus or record exists.

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1 Four meanings of annihilation

Type	Meaning	Possible protection
Global extinction	state becomes identically zero	positive conserved norm/mass/charge
Pattern annihilation	localization or modulation disappears	nonlinear orbital stabil- ity/topology
Coherence annihilation	off-diagonal phases become inaccessible	DFS, dark state, QEC
Identity/sector annihilation	topological charge or attractor is lost	winding gap, pump basin, super- selection

2 Round I: linear strict dynamics

For $U_t = e^{-itA}$,

$$\|U_t\psi\| = \|\psi\|,$$

so a nonzero state never becomes zero. For $P_t = e^{-tA}$,

$$\mathbf{1}^T P_t p = \mathbf{1}^T p, \quad \|P_t(p - u)\| \leq e^{-0.754121 t} \|p - u\|.$$

Heat therefore preserves mass but annihilates the nonuniform pattern. The constant zero mode persists but carries no localization. Finite unitary almost-recurrence is a spectral-torus property, not soliton stability.

Adding uniform loss gives the counterexample

$$\dot{\psi} = -(A + \gamma I)\psi \longrightarrow 0.$$

Thus persistence is channel-dependent and not a property of A alone.

3 Round II: charge, winding and nonlinearity

A phase-invariant Hamiltonian/DNLS has Noether charge

$$Q = \sum_x |\psi_x|^2.$$

If $Q > 0$ is conserved, global zero is impossible. A discrete phase winding

$$W = \frac{1}{2\pi} \sum_x \text{Arg}(\psi_{x+1}/\psi_x)$$

is constant under a continuous homotopy only while $\min_x |\psi_x| > 0$ and no phase increment reaches its cut. Unwinding requires an amplitude-zero or cut event. The linear strict unitary does not prevent component zeros.

DNLS conservation is real, but the repository still lacks a FIN-parameter proof of a localized discrete breather and its orbital stability. Hence “nadsoliton” remains a programme name rather than a completed soliton theorem.

4 Round III: nonequilibrium attractor

For a pumped norm model,

$$\dot{Q} = 2(P - \gamma Q)Q,$$

zero is unstable for $P > 0$ and

$$Q_* = P/\gamma$$

is attracting for every $Q(0) > 0$. This blocks extinction conditionally, but the pump is a resource. Turning it off restores decay. Symmetric pumping also stabilizes a norm/orbit, not one selected D_{12} branch.

Detailed balance normally relaxes excitations unless topology, a conservation law or degeneracy prevents it. Sustained excited persistence is generally nonequilibrium and has an entropy/resource cost.

5 Round IV: quantum and informational protection

Closed unitary dynamics preserves norm/purity; amplitude damping sends excitations to vacuum. A nonzero superselection charge blocks the transition only when vacuum belongs to another charge sector. Dark-state and decoherence-free protection require

$$L_k|\psi\rangle = 0$$

or scalar noise action on a code. Quantum error correction requires

$$PE_a^\dagger E_b P = c_{ab}P.$$

These structures are not supplied by the scalar strict kernel.

Global information conservation does not protect a local pattern: information may disperse into inaccessible correlations or the environment. Autonomous correction also requires an entropy sink. Self-reference or a bootstrap equation is not a stability theorem.

6 Rounds V–VI: falsification and final theorem

The operational library distinguishes Hamiltonian charge, winding protection, pumped attractors and quantum-code protection using norm, inverse participation/localization, coherence, winding and recovery fidelity. Required interventions include pump-off, controlled perturbations, noise scaling, amplitude-zero logging and charge/dark-state tests. A collision test cannot be specified until a multisoliton evolution and charge-balance law exist.

6.1 Final status

Mechanism	Current FIN status	What it protects
Unitary norm	proved	nonzero global state
Heat mass	proved	total probability, not pattern
Wave energy	conditional	phase-space energy
$U(1)$ charge	conditional	global nonzero state
Winding	conditional	topological sector while amplitude gap remains
DNLS breather	open	localized pattern if existence/stability proved
Pump attractor	conditional	nonzero nonequilibrium norm/basin

Mechanism	Current FIN status	What it protects
DFS/dark/QEC	conditional	encoded quantum information
Collision protection	undefined	requires multisoliton theory

Final theorem-scope answer. Current strict FIN does not prove protection of nadsoliton identity. It proves only conservation laws that prevent particular notions of extinction. Full protection requires

A + nonlinear persistent branch + charge/topology or pump + selector + perturbation stability, or a quantum code/dark-state package. None is presently strict-sourced.

7 Recommended next programmes

1. prove or refute a localized DNLS/breather branch for strict A ;
2. interval-certify its Bogoliubov/orbital stability;
3. construct a strict $U(1)$ or winding-charge source and amplitude-gap theorem;
4. define a multisoliton sector and collision charge balance;
5. derive or reject an internal pump/resource law;
6. execute the pump-off/winding/noise persistence protocol with independent records.

A Reproducibility

The authoritative packets run from `FIN_ST912_ST926_Results.json` through `FIN_ST987_ST1001_Results.json`. Six scripts, summaries and the combined test cover all 90 programme packet hashes.

B Nonclaims

No physical immortal object, laboratory persistence, particle annihilation law, photon, gravity, Standard Model or ToE closure is established. All future PDFs remain English.